

Frontend: Theory Assignment 2

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1. What is CSS Box Model?

The CSS Box Model describes how every HTML element is represented as a rectangular box. It consists of four parts:

- Content – the actual text, image, or other content.
- Padding – the space between the content and the border.
- Border – the line surrounding the padding and content.
- Margin – the space outside the border, separating the element from other elements.

The total size of an element depends on these parts and on the value of the box-sizing property.

2. What is media query? Explain with a sample code.

A media query is a CSS technique used to apply different styles depending on conditions such as screen width, height, orientation, or device characteristics. It is an important part of responsive web design because it allows a webpage to adapt to different screen sizes.

```
@media (max-width: 768px) {  
  body {  
    background-color: lightgray;  
  }  
  
  .container {  
    width: 100%;  
    padding: 10px;  
  }  
}
```

3. What are the features of Bootstrap?

Bootstrap is a popular front-end framework for creating responsive and mobile-first websites. Major features include:

- Responsive grid system for different screen sizes.
- Predefined CSS classes and components.
- Buttons, forms, cards, navigation bars, alerts, modals, and other UI components.
- Mobile-first design approach.
- Utility classes for spacing, display, colors, typography, and sizing.
- Consistent design across modern browsers.
- JavaScript-based interactive components.

4. Write an example of basic grid structure in Bootstrap?

Bootstrap's grid system is based on containers, rows, and columns. A basic structure can be written as:

```
<div class="container">  
  <div class="row">  
    <div class="col-md-4">Column 1</div>  
    <div class="col-md-4">Column 2</div>  
    <div class="col-md-4">Column 3</div>
```

```
</div>
</div>
```

5. What is CSS Selectors?

CSS selectors are patterns used to select HTML elements that should receive a particular style. Common selectors include:

- Element selector – selects elements by tag name, e.g. `p { }`.
- Class selector – selects elements with a class, e.g. `.box { }`.
- ID selector – selects an element by ID, e.g. `#header { }`.
- Universal selector – selects all elements, e.g. `* { }`.
- Attribute selector – selects elements based on attributes, e.g. `input[type="text"] { }`.
- Group selector – applies the same style to multiple selectors, e.g. `h1, h2 { }`.
- Descendant/child selectors – select elements according to their relationship in the HTML structure.

6. What are CSS backgrounds, list the properties?

CSS background properties control the background appearance of an HTML element. Important properties are:

- `background-color` – sets the background color.
- `background-image` – sets one or more background images.
- `background-repeat` – controls whether the image repeats.
- `background-position` – sets the starting position of the image.
- `background-size` – controls the size of the background image.
- `background-attachment` – controls whether the background scrolls with the page.
- `background-origin` – specifies the positioning area of the background.
- `background-clip` – specifies how far the background extends.

The shorthand background property can combine several background settings.

7. How can we overlap elements in CSS?

Elements can be overlapped using positioning and the z-index property. An element can be given position: absolute or position: relative, and z-index can be used to control which element appears on top. The parent is commonly given position: relative so that an absolutely positioned child is positioned relative to it.

```
.box1 {
  position: relative;
  z-index: 1;
}

.box2 {
  position: absolute;
  top: 30px;
  left: 30px;
  z-index: 2;
}
```

8. What are the different positioning properties in CSS?

CSS provides five commonly used position values:

- static – the default position; the normal document flow is used.
- relative – the element remains in normal flow but can be shifted using top, right, bottom, or left.
- absolute – the element is removed from normal flow and positioned relative to a suitable positioned ancestor.
- fixed – the element is positioned relative to the viewport and normally stays in the same place while scrolling.
- sticky – the element behaves like relative positioning until a specified scroll position is reached, after which it behaves like a fixed element within its containing context.

9. What is ECMAScript?

ECMAScript is the standardized specification that defines the core features and rules of the JavaScript language. JavaScript is an implementation of ECMAScript, with additional environment-specific features provided by browsers or other runtimes. ECMAScript versions add language features and improvements over time.

10. What is the difference between var, let and const in JavaScript. Explain Temporal Dead Zone?

var, let, and const are used to declare variables in JavaScript.

- var – function-scoped, can be redeclared and reassigned, and is hoisted with an initial value of undefined.
- let – block-scoped, can be reassigned but cannot be redeclared in the same block.
- const – block-scoped, cannot be reassigned or redeclared in the same block. A const variable must be initialized when declared.

Temporal Dead Zone (TDZ):

The TDZ is the period between entering a block and the point where a let or const declaration is initialized. During this period, accessing the variable causes a ReferenceError. Although let and const declarations are hoisted in terms of their binding being created, they are not initialized until execution reaches the declaration.

```
console.log(a); // undefined
var a = 10;

console.log(b); // ReferenceError
let b = 20;

console.log(c); // ReferenceError
const c = 30;
```